Temperley–Lieb algebras

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Notes edited by Grant Barkley

Our goal today is to describe the Temperley–Lieb algebra by diagrams and then interpret these diagrams as maps between representations of $\mathfrak{sl}_2$ and its quantum group.

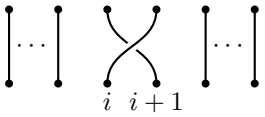
Work over $K = \mathbb{C}(q)$, and put

$$\delta = -(q + q^{-1}).$$

We use this sign convention for the value of a closed loop.

1 Braids and the Hecke algebra

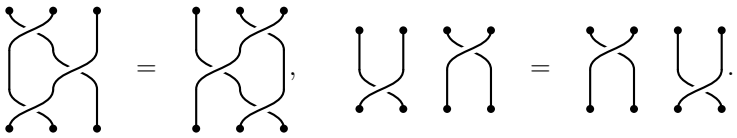
The **braid group** B_k is generated by $T_1, \ldots, T_{k-1}$. The generator T_i crosses strands i and $i+1$, leaving all other strands fixed.

$T_i =$ 

We read diagrams from bottom to top. The relations are

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}, \qquad T_i T_j = T_j T_i \quad \text{if } |i-j| > 1.$$

In pictures,



If we add the relations $T_i^2 = 1$, we get the symmetric group S_k.

Definition 1.1. The **Hecke algebra** $H_k(q)$ is the quotient of $K[B_k]$ by the relations

$$T_i^2 = (q - q^{-1}) T_i + 1.$$

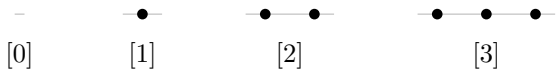
Equivalently, $(T_i - q)(T_i + q^{-1}) = 0$.

At $q = 1$, the corresponding specialization is $\mathbb{C}[S_k]$.

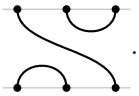
2 The Temperley–Lieb category

Definition 2.1. The **Temperley–Lieb category** $\mathrm{TL}(\delta)$ has objects $[n]$ for $n \geq 0$. A basis of $\mathrm{Hom}([m], [n])$ consists of planar pairings of m points on the bottom of a rectangle and n points on the top, up to isotopy fixing the boundary. The arcs do not cross, and there are no closed components in a basis diagram.

For example, the objects $[0], [1], [2], [3]$ are represented by



Here [0] is the empty set of points. An example of a morphism $[3] \to [3]$ is



If $m+n$ is odd, then $\mathrm{Hom}([m],[n]) = 0$.

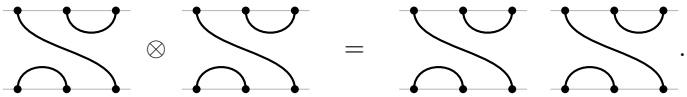
The tensor product is given by placing diagrams side by side. Thus

$$[m] \otimes [n] = [m+n],$$

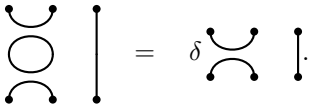
and if $f : [m] \to [p]$ and $g : [n] \to [r]$, then

$$f \otimes g : [m+n] \to [p+r].$$

For example,



Composition is given by stacking diagrams and removing each closed loop at the cost of a factor of δ. For example,



Every diagram is built from the empty diagram, a strand, a cup, and a cap:

$$\mathrm{Id}_{[0]}, \qquad \text{  }, \qquad \text{  } : [0] \to [2], \qquad \text{  } : [2] \to [0].$$

The relations are the two straightening relations and the loop relation:



Writing cup and cap for the two maps, the straightening relations say

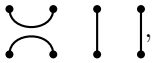
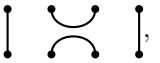
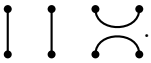
$$(\mathrm{Id} \otimes \mathrm{cap})(\mathrm{cup} \otimes \mathrm{Id}) = \mathrm{Id},$$
$$(\mathrm{cap} \otimes \mathrm{Id})(\mathrm{Id} \otimes \mathrm{cup}) = \mathrm{Id}\,.$$

2.1 The Temperley–Lieb algebra

Definition 2.2. The **Temperley–Lieb algebra** on k strands is

$$\mathrm{TL}_k(\delta) = \mathrm{End}_{\mathrm{TL}(\delta)}([k]).$$

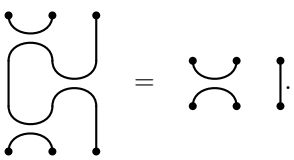
It is generated by $e_1, \ldots, e_{k-1}$, where e_i joins positions i and $i+1$ at both ends and leaves the other strands fixed. For example, in $\mathrm{TL}_4(\delta)$,

$$e_1 = \text{  }, \qquad e_2 = \text{  }, \qquad e_3 = \text{  }.$$

The generators satisfy

$$e_i^2 = \delta e_i, \qquad e_i e_{i\pm1} e_i = e_i, \qquad e_i e_j = e_j e_i \quad \text{if } |i-j| > 1.$$

The middle relation follows by straightening:



Definition 2.3. A **link pattern** on k points is a planar pairing of the points. Let

$$P_k = \mathrm{Hom}_{\mathrm{TL}(\delta)}([0],[k])$$

be the vector space with basis the link patterns.

For example, P_4 has basis



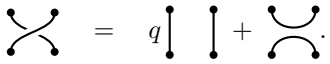
The algebra $\mathrm{TL}_k(\delta)$ acts on P_k by composition. With our convention, this is a left action: $a \cdot p = a \circ p$.

2.2 The map from the Hecke algebra

Define

$$T_i = q\,\mathrm{Id} + e_i.$$

In pictures, this is the rule



Then

$$\begin{aligned} T_i^2 &= q^2\,\mathrm{Id} + 2qe_i + e_i^2 \\ &= q^2\,\mathrm{Id} + (q - q^{-1})e_i \\ &= (q - q^{-1})(q\,\mathrm{Id} + e_i) + \mathrm{Id} \\ &= (q - q^{-1})T_i + \mathrm{Id}\,. \end{aligned}$$

The Temperley–Lieb relations also give the braid relations. Hence we have a surjection

$$H_k(q) \twoheadrightarrow \mathrm{TL}_k(\delta).$$

It is surjective because $e_i = T_i - q\,\mathrm{Id}$.

Remark 2.4. If we instead use generators $f_i = -e_i$, then $f_i^2 = (q + q^{-1})f_i$ and $T_i = q\,\mathrm{Id} - f_i$. This is the positive-parameter convention for the Temperley–Lieb algebra.

3 Representations of $\mathfrak{sl}_2$

Recall that $\mathfrak{sl}_2$ is the Lie algebra of traceless 2×2 matrices. It has generators e, f, h, with relations

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

The universal enveloping algebra $U(\mathfrak{sl}_2)$ is the associative algebra generated by e, f, h subject to these relations.

It is a Hopf algebra, with coproduct

$$\Delta(x) = x \otimes 1 + 1 \otimes x \qquad (x = e, f, h).$$

This gives an action on a tensor product:

$$x(v \otimes w) = xv \otimes w + v \otimes xw.$$

Since the coproduct is cocommutative, the switch map

$$V \otimes W \longrightarrow W \otimes V, \qquad v \otimes w \longmapsto w \otimes v$$

commutes with the $\mathfrak{sl}_2$ action.

There is a q-deformation $U_q(\mathfrak{sl}_2)$, and likewise a quantum group $U_q(\mathfrak{gl}_2)$. The coproduct of $U_q(\mathfrak{sl}_2)$ is not cocommutative. The switch map is replaced by an intertwiner coming from the R-matrix.

3.1 Irreducible representations and paths

The irreducible polynomial representations of GL_2 are indexed by partitions

$$\lambda = (\lambda_1, \lambda_2), \qquad \lambda_1 \geq \lambda_2 \geq 0.$$

Write V_λ for the corresponding representation. The standard representation is $V_{(1,0)} = \mathbb{C}^2$.

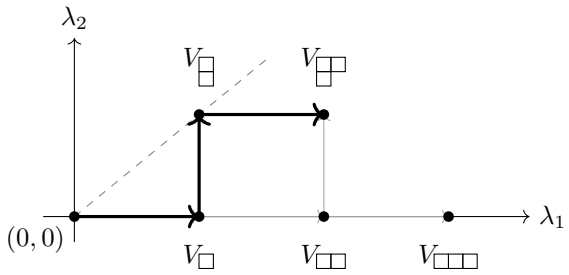
Tensoring with $\mathbb{C}^2$ adds a box:

$$V_\lambda \otimes \mathbb{C}^2 \cong \bigoplus_{\substack{\mu/\lambda = \text{one box} \\ \mu \text{ has at most two rows}}} V_\mu.$$

For example,

$$V_{\ydiagram{1}} \otimes V_{\ydiagram{2}} \cong V_{\ydiagram{3}} \oplus V_{\ydiagram{2,1}}.$$

We can draw the possible additions as paths in the region $\lambda_1 \geq \lambda_2 \geq 0$:



Thus

$$(\mathbb{C}^2)^{\otimes k} \cong \bigoplus_{\substack{\lambda \vdash k \\ \lambda \text{ has at most two rows}}} V_\lambda^{\oplus m_\lambda},$$

where m_λ is the number of paths from $(0,0)$ to λ, taking steps $(1,0)$ or $(0,1)$ and staying in this region.

On restriction to $\mathfrak{sl}_2$, the representation $V_{(\lambda_1,\lambda_2)}$ depends only on $\lambda_1 - \lambda_2$. Write L_r for the irreducible $\mathfrak{sl}_2$ representation of highest weight r, so $\dim L_r = r+1$. Then

$$(\mathbb{C}^2)^{\otimes k} \cong \bigoplus_{\substack{0 \leq r \leq k \\ r \equiv k \pmod 2}} L_r^{\oplus m_{k,r}},$$

where $m_{k,r}$ counts paths from $(0,0)$ to $\left(\frac{k+r}{2}, \frac{k-r}{2}\right)$ in the same region. Equivalently, it counts paths of k up or down steps, starting at height 0, ending at height r, and never going below 0.

Schur's lemma gives

$$\mathrm{Hom}_{\mathfrak{sl}_2}(L_r, L_s) = \begin{cases} \mathbb{C} & r = s, \\ 0 & r \neq s. \end{cases}$$

For generic q, the finite-dimensional type 1 representations of $U_q(\mathfrak{sl}_2)$ have the same tensor product rules. We also write L_r for the corresponding quantum group representation.

4 Cups and caps as intertwiners

Let $V = K^2$ be the standard representation of $U_q(\mathfrak{sl}_2)$, with basis v_1, v_2.

We have

$$V \otimes V \cong L_2 \oplus K,$$

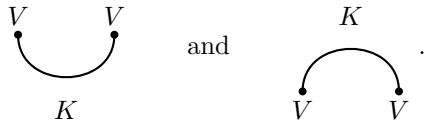
where K on the right is the trivial representation. In particular,

$$\dim \mathrm{Hom}_{U_q(\mathfrak{sl}_2)}(K, V \otimes V) = \dim \mathrm{Hom}_{U_q(\mathfrak{sl}_2)}(V \otimes V, K) = 1.$$

Choose the following maps:

$$\begin{aligned} \mathrm{cup} : K \longrightarrow V \otimes V, \qquad & 1 \longmapsto v_1 \otimes v_2 - q v_2 \otimes v_1, \\ \mathrm{cap} : V \otimes V \longrightarrow K, \qquad & v_1 \otimes v_2 \longmapsto -q^{-1}, \\ & v_2 \otimes v_1 \longmapsto 1, \\ & v_1 \otimes v_1,\ v_2 \otimes v_2 \longmapsto 0. \end{aligned}$$

These are $U_q(\mathfrak{sl}_2)$ intertwiners. They are represented by



The normalizations give

$$\mathrm{cap} \circ \mathrm{cup} = -(q + q^{-1}) = \delta.$$

They also give the two straightening relations

$$\begin{aligned} (\mathrm{Id}_V \otimes \mathrm{cap})(\mathrm{cup} \otimes \mathrm{Id}_V) &= \mathrm{Id}_V, \\ (\mathrm{cap} \otimes \mathrm{Id}_V)(\mathrm{Id}_V \otimes \mathrm{cup}) &= \mathrm{Id}_V . \end{aligned}$$

Thus a closed loop acts by δ, and a bent strand straightens to the identity.

The image of e_i on $V^{\otimes k}$ is

$$\mathrm{Id}_V^{\otimes(i-1)} \otimes (\mathrm{cup} \circ \mathrm{cap}) \otimes \mathrm{Id}_V^{\otimes(k-i-1)} .$$

The braid generator acts by $T_i = q\,\mathrm{Id} + e_i$. At $q = 1$, it switches tensor factors i and $i+1$.

Theorem 4.1. *The assignment $[n] \mapsto V^{\otimes n}$, together with the cup and cap maps above, gives a monoidal equivalence from $\mathrm{TL}(\delta)$ to the full subcategory of $U_q(\mathfrak{sl}_2)$ representations whose objects are the tensor powers of V. In particular,*

$$\mathrm{Hom}_{\mathrm{TL}(\delta)}([m],[n]) \xrightarrow{\sim} \mathrm{Hom}_{U_q(\mathfrak{sl}_2)}(V^{\otimes m}, V^{\otimes n}).$$

Taking $m = n$ gives

$$\mathrm{TL}_n\big(-(q+q^{-1})\big) \cong \mathrm{End}_{U_q(\mathfrak{sl}_2)}(V^{\otimes n}).$$

At $q = 1$, we get

$$\mathrm{TL}_n(-2) \cong \mathrm{End}_{\mathfrak{sl}_2}\big((\mathbb{C}^2)^{\otimes n}\big).$$

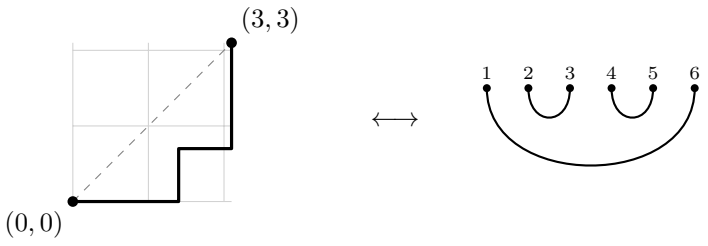
In the positive-parameter convention, this algebra is also written as $\mathrm{TL}_n(2)$.

Taking $m = 0$ identifies link patterns with invariant tensors:

$$P_{2n} \cong \mathrm{Hom}_{U_q(\mathfrak{sl}_2)}(K, V^{\otimes 2n}).$$

The multiplicity of the trivial representation is the number of paths from $(0,0)$ to (n,n) that stay in $\lambda_1 \geq \lambda_2$. These are Dyck paths.

For example, pair an east step with a north step by matching each north step to the most recent unmatched east step:



Thus the link-pattern basis and the path count give

$$\dim P_{2n} = \dim \mathrm{Hom}_{U_q(\mathfrak{sl}_2)}(K, V^{\otimes 2n}) = \frac{1}{n+1}\binom{2n}{n}.$$

AI usage

AI was used for writing these notes based on the speaker's handwritten notes.